

EDUCATION

Degree	Specialization	Institute	Year	CPI
B.Tech	Computer Science & Engineering	Amrita Bangalore	2025	8.01/10
12th CBSE	Physics, Chemistry, & Mathematics	Narayana College	2021	91.0%
10th ICSE	-	New Baldwins	2019	91.6%

WORK EXPERIENCE

- **Software Developer & DevOps Engineer** [Simnovus] Feb 2025 – Present
 - Developing and maintaining tools for large-scale **4G/5G telecom networking simulators**. Automating **simulator deployments, VM creation, and infrastructure management** using **Jenkins, Docker, Linux, and Shell scripting**. Working on **CI/CD pipelines, virtualization environments, system configuration, and simulator orchestration** while also contributing to internal tools and dashboards using **React, Angular, and GoLang**.
- **Machine Learning Intern** [Infosys Springboard] Oct 2024 – Dec 2024
 - Developed a **neural network-based multilingual translation system** supporting six languages using **Python, NLP techniques, and deep learning models**. Built an interactive **Streamlit web application** for real-time translation, implementing data preprocessing pipelines and pretrained language models while coordinating development tasks and presenting the final system.
- **Web Development Intern** [Zima Blue Management] May 2024 – Jul 2024
 - Led a **team of three developers** to design and deploy a responsive corporate website with optimized **UI/UX**. Developed frontend components using **HTML, CSS, and JavaScript** and deployed the application using **Firebase hosting**, ensuring scalable deployment, efficient performance, and maintainable architecture.

PROJECTS

- **Real-Time System Resource Monitoring Dashboard** Jun-Jul 2025
 - Built a **system monitoring dashboard** to track CPU, memory, and network utilization across Linux systems. Developed backend monitoring scripts using **Python and Shell scripting** and visualized system metrics using **web-based dashboards**. Designed the system for **real-time monitoring and performance diagnostics**.
- **Road Safety and Accident Prevention using RF in Speed Limit Zones** Sep-Oct 2024
 - Designed a **vehicle speed control and accident prevention system** using **Arduino Uno, RF transmitters/receivers, ultrasonic sensors, and embedded C**. The system automatically enforces **speed limits in restricted zones such as schools and hospitals** and detects obstacles to trigger emergency braking for enhanced road safety.
- **Secure Multi Client-Server Communication using SSL and Hybrid RSA-AES** Mar-Apr 2024
 - Developed a **secure client-server communication framework** using **SSL/TLS with hybrid RSA-AES encryption**. Implemented secure **key exchange, digital signatures, and encrypted message transmission** to ensure confidentiality, integrity, and authentication. Built using **Python, socket programming, and cryptography libraries** to simulate secure communication between distributed network clients.
- **Genetic Algorithm for Intelligent Vehicle Routing in Logistics** Jun-Aug 2023
 - Implemented a **genetic algorithm based optimization system** to solve the **Vehicle Routing Problem (VRP)** for logistics networks. Developed using **Python and evolutionary optimization techniques** to compute optimal delivery routes based on **distance, demand, travel time, and geographic coordinates**, reducing routing cost and improving delivery efficiency.

PUBLICATIONS

- **Multi-Client Server Based Quantum Key Distribution** [ICCCNT, IIT Indore] Jul 2025
 - **Presented at IIT Indore**, Designed a quantum-secure communication framework for distributed systems using **Quantum Key Distribution (QKD)** to enable scalable and secure key exchange between multiple clients and servers.
- **Cloud-Based Real-Time Anomaly Detection in Network Traffic** [ICOCT, Bengaluru] Jun 2025

- Implemented a **machine learning-based anomaly detection system** for network traffic monitoring using cloud infrastructure. Built real-time analytics pipelines for detecting suspicious activities in distributed network environments.
- **CloudShare: Passwordless Cloud-Based File Storage & Sharing Framework** [ICOCT, Bengaluru] Jun 2025
 - Developed a **secure cloud storage framework** eliminating traditional password authentication using **asymmetric cryptography and challenge-response protocols**. Integrated secure file sharing with **AES encryption and controlled access mechanisms**.
- **Development of a Robust Emergency Communication System Utilizing LoRa and GPS with Multi-Hop Routing** [ICSSAS] Oct 2024
 - Proposed a **long-range emergency communication system** using **LoRa modules and GPS-based location tracking**. Implemented a **multi-hop mesh routing architecture** for transmitting distress signals in disaster scenarios where conventional communication infrastructure fails.
- **License Plate Detection and Recognition Using YOLOv8 and OCR** [15th ICCCNT, IIT Mandi] Jun 2024
 - **Presented at IIT Mandi**, Developed a computer vision based **automatic license plate recognition system** using **YOLOv8 object detection** and evaluated multiple OCR engines including EasyOCR, PaddleOCR, and Tesseract for character extraction.
- **Improvising Energy Efficiency of Heterogeneous Servers using AWS Services and Machine Learning Approaches** [IEEE ICCCNT] Jun 2024
 - Proposed a **machine learning-based approach** to optimize energy consumption in cloud infrastructure. Implemented regression models and deployed containerized workloads using **AWS EC2 and Docker** to improve resource allocation efficiency.
- **Crop Damage Prediction using Machine Learning Approaches** [RAICS] Dec 2023
 - Developed a **machine learning model for agricultural crop damage prediction** using classification algorithms such as Gradient Boosting and Neural Networks to improve accuracy in crop health analysis.

PATENT WORKS

- **Pothole Detection and Repair System and Method Thereof** [Indian Patent] 2025
 - Developed an **automated road maintenance system** capable of detecting potholes using image acquisition and processing techniques. Integrated **computer vision, geolocation tracking, and automated filler dispensing mechanisms** to identify pothole dimensions and perform real-time repairs.
- **IoT-Enabled Pharmaceutical Inventory Management System for Hospital Pharmacies** [Indian Patent] 2024
 - Designed an **IoT-based smart inventory system** to automate pharmaceutical stock management in hospital pharmacies. Implemented **barcode scanning, real-time inventory tracking, automated re-stocking alerts, and expiry monitoring** using **ESP32 microcontrollers, PHP, MySQL, and web-based dashboards** to reduce manual errors and improve operational efficiency.

CERTIFICATIONS AND EXTRA-CURRICULAR ACHIEVEMENTS/ACTIVITIES

- **AWS Academy Graduate – Cloud Architecting** [AWS Academy] Jan 2025
- **AWS Academy Graduate – Cloud Security Foundations** [AWS Academy] Jan 2025
- **Artificial Neural Networks with Keras in Python and R Certificate** [Udemy] Jan 2025
- **Data Manipulation in Python: NumPy and Pandas Certificate** [Udemy] Jan 2025
- **AWS Academy Graduate – Cloud Foundations** [AWS Academy] Dec 2024
- **Recon for Bug Bounty, Penetration Testers and Ethical Hackers Certificate** [Udemy] Aug 2024
- **Applied Generative AI and Natural Language Processing Certificate** [Udemy] Jul 2024
- **Machine Learning Course with Python Certificate** [Udemy] May 2024
- Participated in **Live in Labs**, solving real-world socio-economic problems in a rural village. Jan 2024
- **Workshop on Emerging Research Directions in Computer Vision** Nov 2023
- **DevOps Workshop** covering CI/CD pipelines, containerization, and automation tools. Oct 2023
- Office Bearer of **The EPIC Club**, the videography club at Amrita Vishwa Vidyapeetham. Jan 2023 - Jan 2024
- **Web Developer Certificate** [Acmegrade] Apr 2022
- Volunteered in **AYUDH** social service initiative at Amrita Hospital, **Faridabad**. Aug 2022
- Earned **Black Belt in Karate** specializing in the **Shitō-ryū** style. Jan 2020

SKILLS

- **Programming Languages:** Python, Go, Java, C, C++, JavaScript, Shell Scripting, SQL
- **Web & Frameworks:** React, Angular, HTML, CSS, Flask, Streamlit
- **Cloud & DevOps:** AWS, Docker, Jenkins, CI/CD Pipelines, Virtualization, Linux System Administration
- **Machine Learning & Data:** Scikit-learn, TensorFlow/Keras, NumPy, Pandas, NLP, Computer Vision
- **Networking & Security:** TCP/IP, SSL/TLS, Cryptography (RSA, AES), Network Simulation, Distributed Systems
- **IoT & Embedded Systems:** ESP32, Arduino, Wireless Communication, Raspberry Pi, Sensor Integration
- **Tools:** Git, Jupyter Notebook, VS Code, LaTeX, MySQL, Firebase, Figma
- **Soft Skills:** Leadership, Technical Presentation, Team Collaboration, Problem Solving
- **Languages:** English, Telugu, Kannada, Hindi, Tamil, German (Beginner)